

MTH203: Sequences and series
Lecture 04

1. LEAST UPPER BOUNDS AND GREATEST LOWER BOUNDS

In the last lecture, we defined the notion of a least upper bound. Given a subset T of a partially ordered set S , a *least upper bound* of T is an upper bound s_0 of T such that if s is any other upper bound of T , then $s_0 \leq s$. We also noted that we can analogously define the notion of *greatest lower bound*.

First, let us note the following obvious fact:

Lemma 1. *Let S be a partially ordered set and let $T \subset S$. If α and β are least upper bounds of T , then $\alpha = \beta$.*

Proof. By definition, both α and β are upper bounds of T . Since they are both least upper bounds, we have $\alpha \leq \beta$ and $\beta \leq \alpha$. Due to the antisymmetry property of order relations, we conclude that $\alpha = \beta$. $\square$

Thus, least upper bounds, if they exist, are unique. Similarly, greatest lower bounds, if they exist, are unique. Thus, we may speak of *the* least upper bound of a set. The least upper bound of a set T is also called the *supremum* of T and is denoted by $\sup T$. The greatest lower bound of a set T is also called the *infimum* of T and is denoted by $\inf T$.

We also defined the following property:

Definition 2. Let S be a partially ordered set. We say that S has the *least upper bound property* (*LUB property*) if every *non-empty*, bounded above subset of S has a least upper bound.

Remark 3. We note that it is important to restrict ourselves to *non-empty*, bounded above subsets of S . Indeed, the empty set $\emptyset$, which is a subset of S is trivially seen to be bounded above. Indeed, any element of S is greater than or equal to *any* element of $\emptyset$, for the rather trivial reason that $\emptyset$ does not have any elements. Thus, a least upper bound of $\emptyset$ would have to be a lower bound for the whole of S . Thus, we see that unless we restrict ourselves to non-empty subsets of S , we will end up with a far too restrictive (and useless) notion.

It is natural to define the corresponding notion for lower bounds:

Definition 4. Let S be a partially ordered set. We say that S has the *greatest lower bound property* (*GLB property*) if every *non-empty* bounded below subset of S has a greatest lower bound.

However, it turns out that these two notions are actually the same.

Theorem 5. *Let S be a partially ordered set. Then, S has the GLB property if and only if it has the LUB property.*

Proof. Suppose S has the LUB property. We will now prove that it has the GLB property.

Let T be a non-empty, bounded below subset of S . Let B be the set of lower bounds of T . In other words,

$$B := \{x \in S \mid x \leq t \text{ for all } t \in T\}.$$

The assumption that T is bounded below implies that B is non-empty.

Note that B is bounded above. Indeed, every element of T is greater than every element of B and hence is an upper bound for B . Thus, B has a least upper bound. Let α be a least upper bound of B . We claim that α is a greatest lower bound of T .

First, let us verify that α is a lower bound of T . Let t be any element of T . As we saw above, every element of T is an upper bound for B . Thus, t is an upper bound for B . As α is the least upper bound for B , it follows that $\alpha \leq t$. Thus, we see that α is a lower bound for T .

Now, we show that α is a greatest lower bound of T . Suppose that there is another lower bound β of T . Then $\beta \in B$ by the definition of B . As α is an upper bound for B , we see that $\alpha \geq \beta$. Thus, we see that α is a greatest lower bound of T .

Thus, we have proved that any partially ordered set with the LUB property has the GLB property. The converse can be proved similarly. (*Remark:* We could also just derive the converse from the statement that we have proved by just using the “reverse” order. Do you see this?) $\square$

Thus, the LUB property and the GLB property are logically equivalent and so we only need to talk about one of them.

2. THE REAL NUMBERS

We now define the field of *real numbers* in terms of the field of rational numbers. We will define it to be an ordered field with certain properties that we will specify.

- (R1) $\mathbb{Q}$ is a subset of $\mathbb{R}$.
- (R2) The addition and multiplication operations on $\mathbb{R}$ extend the addition and multiplication operations on $\mathbb{Q}$.

These two properties are expressed by saying that $\mathbb{Q}$ is a *subfield* of $\mathbb{R}$. (Note that these imply that the 0 element of $\mathbb{Q}$ is the same as the 0 element of $\mathbb{R}$. Similarly, the 1 element of $\mathbb{Q}$ is the same as the 1 element of $\mathbb{R}$.)

- (R3) The order relation on $\mathbb{R}$ extends the order relation on $\mathbb{Q}$.
- (R4) $\mathbb{R}$ has the LUB property. (This is called the *Axiom of Completeness*.)

We define $\mathbb{R}$ to be an ordered field satisfying the properties R1-R4. We will *assume* that such a field exists.

An assumption like this should make you uncomfortable. This raises two questions:

- (1) Is the existence of such a field necessarily an assumption (i.e. an axiom) or can it be proved?
- (2) Is there only one such field?

We answer these two questions briefly.

The existence of an ordered field with the listed properties can actually be proved explicitly. It is not a difficult thing to do, but it is time-consuming. Not only does one have to explicitly describe the construction of the field, one also has to then verify that it has the required properties. This is a good educational exercise, but it will not necessarily shed much light on what we are trying to learn in this course. We are more interested in how the real numbers behave rather than what they *are*.

There are actually multiple ways to construct an ordered field with the listed properties and so such an ordered field is certainly not unique. However, any two such ordered fields are *essentially* the same in the sense that we can establish a bijection between them that preserves the algebraic operations as well as the order relation. (We say that the two ordered fields are *isomorphic*.)

Example 6. As a quick application, we will show that the field $\mathbb{R}$ contains an element such that its square is 2.

Let us define T to be the subset of $\mathbb{R}$ with the following property:

$$T = \{x \in \mathbb{Q} \mid x^2 < 2\}.$$

We proved in Lecture 3 that this set is bounded above. We also saw that it does not have a least upper bound in $\mathbb{Q}$. However, by the Axiom of Completeness, it does have a least upper bound in $\mathbb{R}$. Let us denote this least upper bound by α .

In the following proof, we will use an important property of the real numbers called the Archimedean property. This is the statement that *given any real number x , there exists a natural number $n > x$* . This statement will be proved in Lecture 5 as a consequence of the Axiom of Completeness. (Note that we have used the analogous statement for the rational numbers in Example 11 of Lecture 3. However, the Archimedean property for $\mathbb{Q}$ is very easy to verify.)

Since $0 \in T$, we have $\alpha > 0$. We claim that $\alpha^2 = 2$. Indeed, suppose $\alpha^2 < 2$. Then, we can obtain a contradiction as in Example 11 of Lecture 3. Indeed, we have $2 - \alpha^2 > 0$ and so $(2\alpha + 1)/(2 - \alpha^2) > 0$. We choose n to be any natural number greater than $(2\alpha + 1)/(2 - \alpha^2)$ so that $(2\alpha + 1)/n > 2 - \alpha^2$. We set $\alpha_1 := \alpha + 1/n$. Then,

$$\begin{aligned}\alpha_1^2 &= \alpha^2 + 2\alpha/n + 1/n^2 \\ &= 2 + \left[\frac{2\alpha + 1/n}{n} - (2 - \alpha^2) \right] \\ &\geq 2 + \left[\frac{2\alpha + 1}{n} - (2 - \alpha^2) \right] \\ &> 2.\end{aligned}$$

Thus, $\alpha_1 \in T$, but $\alpha_1 > \alpha$, which gives us a contradiction since α is an upper bound for T . Thus, we conclude that $\alpha^2 \geq 2$.

If $\alpha^2 > 2$, we can again get a contradiction by a similar argument. We choose n to be a natural number greater than $2\alpha/(\alpha^2 - 2)$ and set $\alpha_1 = \alpha - 1/n$. Then, a similar calculation to the one above shows that $\alpha_1^2 > 2$. One can show also easily verify that $\alpha_1 > 0$. (See Example 11 in Lecture 3 for the detailed argument.) Using Lemma 12 in Lecture 3, we can conclude that α_1 is also an upper bound of T , which once again gives us a contradiction.

Thus, we must have $\alpha^2 = 2$. We will denote this element α by $\sqrt{2}$.

3. ANOTHER DESCRIPTION OF THE SUPREMUM

We have the following convenient characterization of supremums of subsets of $\mathbb{R}$.

Lemma 7. *Let $S \subset \mathbb{R}$. Then $\alpha = \sup S$ if and only if the following two conditions hold:*

- (a) α is an upper bound for S .
- (b) For every $\epsilon > 0$, there exists an element $s \in S$ such that $s > \alpha - \epsilon$.

Proof. Suppose $\alpha = \sup S$. Then (a) obviously holds. If for some ϵ , there does not exist any $s \in S$ such that $s > \alpha - \epsilon$, then $\alpha - \epsilon$ is an upper bound for S . This contradicts the assumption that $\alpha = \sup S$. This, (b) also must hold.

Now, suppose (a) and (b) are true. We only have to show that if any real number β is less than α , then β cannot be an upper bound of S . Given any $\beta < \alpha$, let us set $\epsilon = \alpha - \beta$ so that $\alpha - \epsilon = \beta$. Then, by (b), there exists some $s \in S$ such that $\beta < s$. Thus, β is not an upper bound of S . This completes the proof. $\square$

4. READING SUGGESTIONS

This lecture corresponds to Section 1.3 from the text ([?]). The computation in Example ?? is essentially the same as the one in Theorem 1.4.5 of the text.

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.